

$$\cancel{V(s_t, I_t, t) = \left[\frac{I_t}{s} \right] \cdot H(x, t)}, \quad (x) = \left[\frac{I_t}{s} \right] \leftarrow$$

$$\frac{\partial^2 V}{\partial s^2} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial x^2} + r s \frac{\partial V}{\partial s} + \underbrace{\frac{\partial V}{\partial t}}_{\text{red}} - rV = 0$$

$$\frac{\partial V}{\partial s} = \frac{I_t}{s}, \quad \frac{\partial V}{\partial x} = \frac{I_t}{s} \cdot s = \frac{I_t}{s} \cdot s = I_t$$

$$\frac{\partial V}{\partial s} = \frac{I_t}{s} \cdot s = I_t$$

$$\frac{\partial^2 V}{\partial s^2} = \frac{\partial}{\partial s} \left(\frac{I_t}{s} \right) = -\frac{I_t}{s^2}$$

$$= \left[\frac{\partial^2 H}{\partial x^2} \cdot \frac{I_t}{s} + \frac{\partial H}{\partial x} \cdot \left(-\frac{I_t}{s^2} \right) \right]$$

$$= \frac{1}{s} x^2 \frac{\partial^2 H}{\partial x^2}$$

$$\cancel{s} \frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \cancel{s}^2 \left(\cancel{s} x^2 \frac{\partial^2 H}{\partial x^2} \right) + \cancel{r} \cancel{s} \left(H - x \frac{\partial H}{\partial x} \right) + \cancel{s} \frac{\partial H}{\partial x} - \cancel{r} \cancel{s} H = 0$$

$$\Rightarrow \frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 H}{\partial x^2} + \frac{\partial H}{\partial x} (1 - rx) = 0 \quad H(x, t)$$

$$\underline{\underline{H}} \quad , \quad \underline{\underline{x}}$$